AI-Powered Agriculture Crop Advisory Assistant

REST API Documentation

Base URL: http://localhost:8000 | Interactive Docs: http://localhost:8000/docs

GET /health

System health check — backend status, index loading, ML model readiness

Request: None

Response: {"status":"healthy","version":"1.0.0","vector_store_cases":1500,"suitability_model_loaded":true}

POST /api/query

Full 8-agent advisory pipeline — the main endpoint

Request: {"query":"string","top_k":5,"filters":{"state":"Andhra Pradesh","season":"Kharif"}}

Response: {"recommended_crops":[...],"similar_cases":[...],"yield_prediction":{...},"risk_analysis":{...},"validation":{...}}

POST /api/search

Hybrid dense+sparse search over historical cases

Request: {"query":"string","top_k":10,"state":"Maharashtra","season":"Kharif","crop":"Rice"}

Response: {"total_retrieved":10,"cases":[{"case_id":"CASE_0001","crop":"Rice","yield":3.65,"rrf_score":0.033}]}

POST /api/recommendation

Standalone crop suitability ranking

Request: {"state":"Punjab","season":"Rabi","ph":7.0,"n":120,"p":60,"k":40,"rainfall":150,"temperature":18}

Response: {"candidates":[{"crop":"Wheat","suitability_probability":0.95,"historical_avg_yield":4.62,"risk_level":"LOW"}]}

POST /api/prediction

ML yield prediction with 80% confidence interval

Request: {"crop":"Cotton","state":"Gujarat","season":"Kharif","area":10,"n":100,"ph":7.4,"rainfall":750,"temperature":29}

Response: {"predicted_yield":2.14,"unit":"tonnes/hectare","expected_range_80_ci":[1.88,2.40],"standard_deviation":0.20}

POST /api/analysis

Soil & climate impact analysis — limiting factor identification

Request: {"crop":"Maize","ph":5.2,"rainfall":450,"temperature":34}

Response: {"limiting_factors":["pH 5.2 is acidic for Maize"],"recommendations":["Apply agricultural lime"]}

POST /api/feedback

Submit user feedback on advisory quality

Request: {"query":"string","rating":5,"feedback_category":"ACCURATE","comments":"string"}

Response: {"status":"success","message":"Feedback recorded","feedback_id":"FB_20261007_001"}

POST /api/evaluate

Run automated evaluation benchmark suite

Request: {}

Response: {"status":"completed","precision_at_5":0.83,"recall_at_5":0.78,"mrr":0.925,"yield_r2":0.9876}

GET /api/evaluation/latest

Retrieve most recent benchmark report

Request: None

Response: Complete evaluation JSON report